

OSLAB PRACTICAL-11 PROGRAMS & OUTPUT

Date: 03-10-26

Program-1 CODE & OUTPUT:

```
Final counter = 200000
sainath@sainath:~/OSLAB/practical_11$ cat program1.c
#include <stdio.h>
#include <pthread.h>

int counter = 0;

void *increment(void *arg) {
    for (int i = 0; i < 100000; i++)
        counter++;

    return NULL;
}

int main() {
    pthread_t t1, t2;

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final counter = %d\n", counter);

    return 0;
}
sainath@sainath:~/OSLAB/practical_11$ ./program1
Final counter = 123378
```

Program-2 CODE & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_11$ gedit program2.c
sainath@Sainath:~/OSLAB/practical_11$ gcc program2.c -o program2
sainath@Sainath:~/OSLAB/practical_11$ cat program2.c
#include <stdio.h>
#include <pthread.h>

int counter = 0;
pthread_mutex_t lock;

void *increment(void *arg) {
    for (int i = 0; i < 100000; i++) {
        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main() {
    pthread_t t1, t2;

    pthread_mutex_init(&lock, NULL);

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final counter = %d\n", counter);

    pthread_mutex_destroy(&lock);

    return 0;
}
sainath@Sainath:~/OSLAB/practical_11$ ./program2
Final counter = 200000
```

History:

```
595 cd OSLAB
596 ls
597 mkdir practical_11
598 cd practical_11
599 gedit program1.c
600 gcc program1.c -o program1
601 ./program1
602 gedit program2.c
603 gcc program2.c -o program2
604 cat program2.c
605 ./program2
606 cat program1.c
607 ./program1
608 history
```